

| Question    |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |     |     |       |       |      |
|-------------|-----|--|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|             |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 6<br>H<br>T | (a) |  |  | <b>Indicative content:</b><br>The term latent is used because the energy supplied does not raise the temperature of the substance but merely goes to enable the change of state to take place [per kg]. The term “fusion” refers to the change from a solid to liquid at its melting point and the energy supplied goes into breaking bonds between molecules. In the case of fusion, only a small number of bonds are broken. The latent heat of vaporisation refers to a liquid changing to a gas and is always a much larger value because the change requires a much larger number of bonds to be broken. | 6               |     |     |       | 0     | 0    |
|             |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                 |     |     |       |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks Available |     |     |       |       |      |
|----------|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  |  |  | <p><b>1-2 marks</b><br/>A limited attempt is made at either the term latent as being explained as supplying energy to a change of state without a change in temperature or a description of the difference in molecular bond breaking is given during a change of state.<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |     |     |       |       |      |

|  |     |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |          |          |          |           |          |          |
|--|-----|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|----------|-----------|----------|----------|
|  | (b) | (i)  | <p>Temp rise = <math>\Delta\theta = 640[^\circ\text{C}]</math> (1)<br/>Energy required to raise temp to boiling point<br/>= <math>mc\Delta\theta = 500 \times 900 \times 640</math> <b>ecf</b> = 288 000 000 [J] (1)<br/>Energy required to melt aluminium = <math>500 \times 400\,000 = 200\,000\,000</math> J (1)<br/>Total energy required = <math>4.88 \times 10^8</math> J (1) <b>ecf</b><br/><br/>Substitution and conversion: time = <math>\frac{4.88 \times 10^8 \text{ecf}}{8.7 \times 10^4}</math><br/><br/>Answer <b>with unit</b> = 5 609.2 s <b>or</b> 93.5 min <b>or</b> 1.56 h (1)<br/>A conclusion consistent with their answer i.e. the one hour deadline is not met (1)<br/><br/><b>Alternative for 5<sup>th</sup> and 6<sup>th</sup> marks:</b><br/>Number of kWh = 87 converted to J = <math>3.13 \times 10^8</math> (1)<br/>This is less than the energy required <b>so</b> the one hour deadline is not met (1)<br/><br/><b>Alternative for 5<sup>th</sup> and 6<sup>th</sup> marks:</b><br/><math>4.88 \times 10^8</math> J = 135.6 [kWh] (1)<br/>This is more energy than the heater can provide in one hour <b>so</b> the one hour deadline is not met (1)<br/><br/><b>Alternative:</b><br/>Energy supplied = <math>87\,000 \times 3\,600</math> (1) = <math>3.13 \times 10^8</math> [J] (1)<br/>Temp rise = <math>\Delta\theta = 640[^\circ\text{C}]</math> (1)<br/><math>3.13 \times 10^8 = m(c\Delta\theta + L)</math> so<br/><math>3.13 \times 10^8 = m(900 \times 640 + 400\,000)</math> (1)<br/><math>m = \frac{3.13 \times 10^8}{976\,000} = 320.7</math> [kg] (1)<br/>A conclusion consistent with their answer i.e. the one hour deadline is not met (1)</p> |          |          |          |           |          |          |
|  |     | (ii) | <p>Energy is needed to raise the temperature of heater and container / some energy is lost [to the surroundings]</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 1        |          |          | 1         |          |          |
|  |     |      | <p><b>Question 6 total</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <b>7</b> | <b>0</b> | <b>6</b> | <b>13</b> | <b>6</b> | <b>0</b> |